BILL OF MATERIALS (BOM)

Obstacle-Avoiding Robot — Physical Build and Tinkercad Simulation

1. Design basis

This BOM covers the two-wheel obstacle-avoiding robot discussed in this project. The intended physical build uses an Arduino Uno, an L298N motor-driver module, two DC geared motors, an HC-SR04 ultrasonic sensor mounted on an SG90-compatible servo, and a 2-cell (2S) 18650 battery pack. The current Tinkercad circuit may use an L293D and a 9 V battery; these are identified separately as simulation choices, not additional physical parts.

2. Electronics and electrical items

Item	Specification / description	Qty.	Purpose / notes
1	Arduino Uno R3 (ATmega328P, 5 V logic)	1	Main controller for sensing and motor control.
2	L298N dual H-bridge motor-driver module	1	Specified for the physical design; drives both DC motors.
3	L293D dual H-bridge IC/module	Simulation only	Use only if this is the driver in the existing Tinkercad circuit; not an extra physical part.
4	HC-SR04 ultrasonic sensor	1	Measures obstacle distance; mounted facing forward on the servo bracket.
5	SG90-compatible 9 g servo	1	Moves the ultrasonic sensor between left, centre and right scan positions.
6	DC geared motors compatible with driver and battery	2	Left and right drive motors; confirm voltage and stall-current ratings.
7	Wheels compatible with motor shafts	2	Drive wheels; hub/shaft fit must match selected motors.
8	Matched 18650 rechargeable Li-ion cells	2	Connected in series as a 2S pack: 7.4 V nominal, 8.4 V fully charged.
9	2-cell series battery holder / enclosure	1	Securely holds the two cells and supports series connection.
10	2S Li-ion BMS/protection board	1	Recommended for cell protection; current rating must suit the load.
11	Main power switch	1	Disconnects battery power for safe handling.
12	Inline fuse and holder	1 set	Select fuse rating based on expected motor current and wiring.
13	Regulated 5 V servo supply, if needed	Optional 1	Use if the Arduino 5 V rail cannot supply servo current reliably; share common ground.
14	Jumper/hookup wires	As needed	Signal, power and ground connections; use suitable gauge for motor current.
15	Screw terminals / connector blocks	As needed	Removable motor, battery and sensor connections.
16	Breadboard or prototyping PCB	1	For prototype interconnections; final PCB may be a conceptual mechanical model.

3. Mechanical parts and mounting hardware

Item	Specification / description	Qty.	Purpose / notes
1	Lower chassis deck	1	Supports motors, wheels, caster and battery; dimensions follow the Fusion 360 model.
2	Upper chassis deck	1	Supports electronics and wiring if using the two-deck design.
3	Standoffs / pillars	As designed	Separate the decks; match their height to the CAD assembly.
4	Motor mounting brackets	2	Secure the geared motors to the chassis.
5	Caster wheel / ball caster	1	Provides a third support point for the two-wheel-drive chassis.
6	Servo bracket / sensor mount	1	Holds the servo and aims the HC-SR04 forward at centre position.
7	Battery holder bracket / straps	1 set	Prevents battery movement during turns.
8	Screws, nuts, washers and standoffs	As needed	Choose sizes to match chassis holes and brackets.
9	Cable ties / wire clips	As needed	Keep wiring clear of wheels and moving servo.

4. Simulation vs. physical build

Tinkercad pin assignments from the supplied prebuilt circuit: HC-SR04 TRIG D12, ECHO D11; motor direction pins D7, D6, D5 and D4; original auxiliary outputs D13 and D10; servo signal D9 when added. If the simulator uses an L293D and a 9 V battery, document these as simulation components. **Physical design:** This BOM specifies an L298N module and a 2S 18650 pack. The final schematic and code must match the actual driver and wiring.

Power and safety: A 2S 18650 pack is 7.4 V nominal and 8.4 V fully charged. Never connect the pack directly to the Arduino 5 V pin. Check motor voltage and stall-current ratings against the driver. The servo may need a separate regulated 5 V supply if the Arduino rail is insufficient; all grounds must be common. Use a suitable charger and a protected battery pack.

Items not priced or assigned exact part numbers: vendor prices, motor model/rating, fuse rating, screw sizes and exact chassis dimensions depend on the final selected components and CAD design. These have not been guessed.